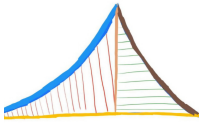
Matrices In Geometry



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. AE 2026

A. Formulae

1.1.1

$$\Pr(M) + \Pr(E) - \Pr(ME) + \Pr(M+E)' = 1 \tag{1.1.1.1}$$

1.1.2

$$A + A' = 1, \quad AA' = 0 \tag{1.1.2.1}$$

$$\implies \Pr(A + A') = \Pr(1) = 1 \quad \Pr(AA') = \Pr(0) = 0 \tag{1.1.2.2}$$

1.1.3

$$\Pr(A + A') = \Pr(A) + \Pr(A') \tag{1.1.3.1}$$

1.1.4 Difference rule of probability

$$\Pr(AB') = \Pr(A) - \Pr(AB) \tag{1.1.4.1}$$

Solution: From (1.1.2.1)

$$A = A \cdot 1 = A(B + B') = AB + AB' \tag{1.1.4.2}$$

Also,

$$(AB)(AB') = 0 \tag{1.1.4.3}$$

From (1.1.3.1),

$$\Pr(A) = \Pr(AB + AB') = \Pr(AB) + \Pr(AB') \tag{1.1.4.4}$$

Rearranging, we get:

$$P(AB') = P(A) - P(AB) \tag{1.1.4.5}$$

1.1.5 The Laplace transform is defined as

$$g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s) = \int_0^\infty e^{-st} g(t)\, dt \tag{1.1.5.1}$$

1.1.6

$$g'(t) \overset{\mathcal{L}}{\longleftrightarrow} sG(s) - g(0) \tag{1.1.6.1}$$

Solution: Applying integration by parts,

$$u = e^{-st}, dv = g'(t)\, dt \tag{1.1.6.2}$$

$$\implies \int_0^\infty e^{-st} g'(t)\, dt = \left[e^{-st} g(t)\right]_0^\infty - \int_0^\infty \left(-se^{-st}\right) g(t)\, dt \tag{1.1.6.3}$$

$$= -g(0) + s\int_0^\infty e^{-st} g(t)\, dt \tag{1.1.6.4}$$

yielding (1.1.6.1).

1.1.7

$$g''(t) \overset{\mathcal{L}}{\longleftrightarrow} s^2 G(s) - sg(0) - g'(0) \tag{1.1.7.1}$$

1.1.8

$$e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s-a) \tag{1.1.8.1}$$

Solution:

$$\implies e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty e^{-st} \left(e^{at} g(t)\right) dt \tag{1.1.8.2}$$

$$= \int_0^\infty e^{-(s-a)t} g(t)\, dt \tag{1.1.8.3}$$

yielding (1.1.8.1).

1.1.9 The unit step function is defined as

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} \tag{1.1.9.1}$$

1.1.10

$$u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s} \tag{1.1.10.1}$$

Solution:

$$\int_0^\infty e^{-st}\, dt = \left(-\frac{e^{-st}}{s}\right)_0^\infty = \frac{1}{s} \quad \Re(s) > 0 \tag{1.1.10.2}$$

1.1.11

$$tu(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s^2} \tag{1.1.11.1}$$

Solution: Differentiating (1.1.10.2) with respect to s results in (1.1.11.1).

1.1.12

$$e^{-at} u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+a}, \quad \Re(s) > \Re(a). \tag{1.1.12.1}$$

Solution: See (1.1.10.1) and (1.1.8.1).

1.1.13

$$\begin{aligned} \cos(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2+a^2} \\ \sin(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2+a^2} \end{aligned} \quad \Re(s) > a \tag{1.1.13.1}$$

Solution: From (1.1.8.1) and (1.1.10.1),

$$e^{\jmath at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s-\jmath a}, \quad \Re(s) > a \tag{1.1.13.2}$$

$$\implies \cos(at)u(t) + \jmath \sin(at)u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s+\jmath a}{(s-\jmath a)(s+\jmath a)} \tag{1.1.13.3}$$

$$= \frac{s}{s^2+a^2} + \jmath \frac{a}{s^2+a^2} \tag{1.1.13.4}$$

resulting in (1.1.13.1).

1.1.14 Solve

$$\frac{d^2y}{dx^2} = -\alpha^2 y, \alpha \neq 0 \tag{1.1.14.1}$$

Solution: Applying the Laplace transform on both sides, and using (1.1.7.1),

$$s^2Y(s) - sy(0) - y'(0) = -\alpha^2 Y(s) \tag{1.1.14.2}$$

Substituting

$$y(0) = \beta, y'(0) = \gamma \tag{1.1.14.3}$$

in (1.1.14.2),

$$Y(s) = \frac{s\beta + \gamma}{s^2+\alpha^2} \tag{1.1.14.4}$$

$$= \frac{s\beta}{s^2+\alpha^2} + \frac{\gamma}{s^2+\alpha^2} \tag{1.1.14.5}$$

yielding

$$y(t) = \beta\cos(\alpha t) + \frac{\gamma}{\alpha}\sin(\alpha t) \tag{1.1.14.6}$$

from (1.1.13.1).

1.1.15 (Euler's Method of Forward Differences.) Numerically solve

$$\frac{dy}{dx} = f(x,y), y(x_0) = y_0 \tag{1.1.15.1}$$

Solution: By definition,

$$\left.\frac{dy}{dx}\right|_{x=x_0} = \lim_{h\to 0} \frac{y(x_0+h) - y(x_0)}{h} \tag{1.1.15.2}$$

which can be expressed using the notation

$$y_0' = \frac{y_1 - y_0}{h} \tag{1.1.15.3}$$

Similarly,

$$y_n'' = \frac{y_{n+1}' - y_n'}{h} \tag{1.1.15.4}$$

From (1.1.15.3),

$$y'_{n+1} = \frac{y_{n+2} - y_{n+1}}{h}, y'_n = \frac{y_{n+1} - y_n}{h} \tag{1.1.15.5}$$

yielding

$$y''_n = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \tag{1.1.15.6}$$

upon simplification.

1.1.16 Kronecker delta function

$$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.16.1}$$

1.1.17

$$f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi n f_0 t}\, dt = \delta_n \tag{1.1.17.1}$$

1.1.18 The Fourier Series. Let

$$x(t) = \sum_{k=-\infty}^{\infty} x_0(t)\,(t - nT_0) \tag{1.1.18.1}$$

be a periodic signal with

$$x(t) = \begin{cases} x_0(t), & t \in \left(-\frac{T_0}{2}, \frac{T_0}{2}\right) \\ 0 & \text{otherwise} \end{cases}. \tag{1.1.18.2}$$

Then,

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{\mathrm{j}2\pi n f_0 t}, \quad f_0 = \frac{1}{T_0} \tag{1.1.18.3}$$

Show that

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.18.4}$$

Solution: From (1.1.18.3),

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} \left(\sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t} \right) e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.18.5}$$

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi (k-n) f_0 t}\, dt \tag{1.1.18.6}$$

$$= \frac{1}{f_0} \sum_{k=-\infty}^{\infty} c_k \delta_{k-n} \tag{1.1.18.7}$$

upon substituting from (1.1.18.2) and (1.1.17.1), yielding (1.1.18.4).

1.1.19 The rectangular function. The unit rectangular function is defined as

$$\text{rect}\,(t) = \begin{cases} 1 & t \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ 0 & \text{otherwise} \end{cases} \tag{1.1.19.1}$$

1.1.20 Square Wave. Consider the Fourier series with

$$x_0(t) = \text{rect}\,(t), \quad T_0 = 2. \tag{1.1.20.1}$$

Show that

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} \tag{1.1.20.2}$$

Solution: Substituting $f_0 = \frac{1}{2}$ in (1.1.18.4),

$$c_n = \frac{1}{2}\int_{-1}^{1} e^{-j\pi kt}\, dt \tag{1.1.20.3}$$

$$= \left[\frac{\sin(\pi kt)}{2\pi k}\right]_{-\frac{1}{2}}^{\frac{1}{2}} \tag{1.1.20.4}$$

The following code verifies the result in Fig. 1.1.20.1. Note that Fig. 1.1.20.1 is continuous while (1.1.20.1) is discontinuous. So (1.1.20.2) provides only an approximation. This is known as the Gibbs' phenomenon.

codes/Fourier.py

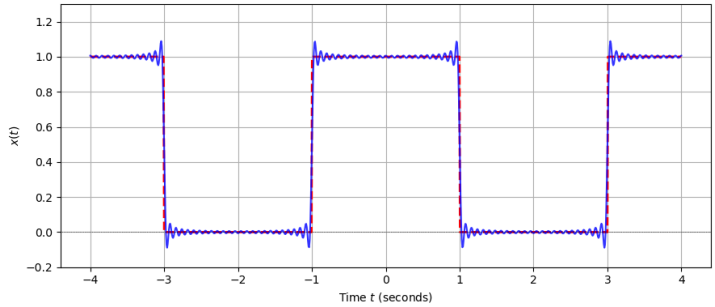


Fig. 1.1.20.1. Gibbs phenomenon

1.1.21 If

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} c_n, \tag{1.1.21.1}$$

show that

$$x(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} c_n e^{-\mathrm{J}2\pi n t_0}, \tag{1.1.21.2}$$

Solution:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0\left(t-t_0\right) e^{-\jmath 2\pi n f_0 t}\, dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0\left(t\right) e^{-\jmath 2\pi n f_0 (t+t_0)}\, dt \tag{1.1.21.3}$$

$$= e^{\jmath 2\pi n f_0 t_0} \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0\left(t\right) e^{-\jmath 2\pi n f_0 t}\, dt \tag{1.1.21.4}$$

after a change of variables. Note that the limits of integration are unchanged due the periodicity of $x(t)$. Transferring the exponential term in (1.1.21.4) to the L.H.S yields (1.1.21.2).

1.1.22 The standard 2-D basis vectors are defined as

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.1.22.1}$$

Also,

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.1.22.2}$$

1.1.23 Find the echelon form of the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.1.23.1}$$

Solution: Following row reduction,

$$\mathbf{A} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{1.1.23.2}$$

1.1.24 The rank of a matrix is defined to be the number of nonzero rows in echelon form. $\therefore \text{rank}\left(\mathbf{A}\right) = 1$.

1.1.25 A matrix is said to be singular if its rank is nonzero.

1.1.26 For

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top, \text{rank}\left(\mathbf{A}\right) = 1. \tag{1.1.26.1}$$

where both $\mathbf{u}$ and $\mathbf{v}$ are n-dimensional real-valued non-zero column vectors.

Solution: Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}. \tag{1.1.26.2}$$

Then

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 4 & 2 \end{pmatrix} \tag{1.1.26.3}$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.1.26.4}$$

From (1.1.23.2) and Definition 1.1.24, we obtain (1.1.26.1).

1.1.27

$$\left|\mathbf{A}^\top\right| = \left|\mathbf{A}\right| \tag{1.1.27.1}$$

1.1.28

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right| \left|\mathbf{B}\right| \tag{1.1.28.1}$$

Solution: If $\mathbf{A}$ is invertible

$$\mathbf{A} = \mathbf{E}_1\mathbf{E}_2 \cdots \mathbf{E}_k \tag{1.1.28.2}$$

where $\mathbf{E}_i$ are invertible matrices. For any single elementary matrix $\mathbf{E}$, it is easy to verify that

$$\left|\mathbf{EB}\right| = \left|\mathbf{E}\right| \left|\mathbf{B}\right|. \tag{1.1.28.3}$$

$$\therefore \left|\mathbf{AB}\right| = \left|\mathbf{E}_1\mathbf{E}_2 \cdots \mathbf{E}_k\mathbf{B}\right| \tag{1.1.28.4}$$

$$= \left|\mathbf{E}_1\right| \left|\mathbf{E}_2 \cdots \mathbf{E}_k\mathbf{B}\right| \tag{1.1.28.5}$$

$$= \left|\mathbf{E}_1\right| \left|\mathbf{E}_2\right| \cdots \left|\mathbf{E}_k\right| \left|\mathbf{B}\right| \tag{1.1.28.6}$$

For $\mathbf{B} = \mathbf{I}$,

$$\left|\mathbf{A}\right| = \left|\mathbf{E}_1\right| \left|\mathbf{E}_2\right| \cdots \left|\mathbf{E}_k\right| \tag{1.1.28.7}$$

which upon substituting in (1.1.28.6) yields

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right| \left|\mathbf{B}\right| \tag{1.1.28.8}$$

1.1.29

$$\left|\mathbf{A^{-1}}\right| = \frac{1}{\left|\mathbf{A}\right|} \tag{1.1.29.1}$$

1.1.30 If

$$\mathbf{Ax} = \lambda\mathbf{x}, \tag{1.1.30.1}$$

then λ is an eigenvalue $\mathbf{x}$ of $\mathbf{A}$.

1.1.31 Let $\lambda_1, \lambda_2, ..., \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then,

$$\sum_{i=1}^{n} \lambda_i = \text{Tr}\left(\mathbf{A}\right) \tag{1.1.31.1}$$

$$\prod_{i=1}^{n} \lambda_i = \left|\mathbf{A}\right| \tag{1.1.31.2}$$

1.1.32 $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.
Solution: From (1.1.29.1),

$$\left|\mathbf{A} - \lambda\mathbf{I}\right| = \left|\left(\mathbf{A} - \lambda\mathbf{I}\right)^\top\right| \tag{1.1.32.1}$$

$$= \left|\mathbf{A}^\top - \lambda\mathbf{I}\right|, \tag{1.1.32.2}$$

completing the proof.

1.1.33 The equation of a conic with directrix $\mathbf{n}^\top\mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\text{g}\left(\mathbf{x}\right) = \mathbf{x}^\top\mathbf{V}\mathbf{x} + 2\mathbf{u}^\top\mathbf{x} + f = 0 \tag{1.1.33.1}$$

where

$$\mathbf{V} = \left\|\mathbf{n}\right\|^2\mathbf{I} - e^2\mathbf{n}\mathbf{n}^\top, \tag{1.1.33.2}$$

$$\mathbf{u} = ce^2\mathbf{n} - \left\|\mathbf{n}\right\|^2\mathbf{F}, \tag{1.1.33.3}$$

$$f = \left\|\mathbf{n}\right\|^2\left\|\mathbf{F}\right\|^2 - c^2e^2 \tag{1.1.33.4}$$

Solution:

$$\left\|\mathbf{x} - \mathbf{F}\right\|^2 = e^2\frac{\left(\mathbf{n}^\top\mathbf{x} - c\right)^2}{\left\|\mathbf{n}\right\|^2} \tag{1.1.33.5}$$

$$\implies \left\|\mathbf{n}\right\|^2\left(\mathbf{x} - \mathbf{F}\right)^\top\left(\mathbf{x} - \mathbf{F}\right) = e^2\left(\mathbf{n}^\top\mathbf{x} - c\right)^2 \tag{1.1.33.6}$$

$$\implies \left\|\mathbf{n}\right\|^2\left(\mathbf{x}^\top\mathbf{x} - 2\mathbf{F}^\top\mathbf{x} + \left\|\mathbf{F}\right\|^2\right) = e^2\left(c^2 + \left(\mathbf{n}^\top\mathbf{x}\right)^2 - 2c\mathbf{n}^\top\mathbf{x}\right) \tag{1.1.33.7}$$

$$= e^2\left(c^2 + \left(\mathbf{x}^\top\mathbf{n}\mathbf{n}^\top\mathbf{x}\right) - 2c\mathbf{n}^\top\mathbf{x}\right) \tag{1.1.33.8}$$

which can be expressed as (1.1.33.1) after simplification.

1.1.34 The polar form of a conic section is given as:

$$r = \frac{ep}{1 + e\cos\left(\theta - \theta_0\right)} \tag{1.1.34.1}$$

where

- r is the distance from the focus(origin) to a point on the curve.
- e is the eccentricity of the conic
- p is the distance from the focus to the directrix
- θ_0 is the angle that defines the orientation of the major axis of the conic section relative to the polar reference axis ($\theta = 0$).

Solution: From (1.1.33.1), the general equation of a conic in matrix form is given

by

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{1.1.34.2}$$

For a conic with a focus at the origin ($\mathbf{F} = \mathbf{0}$), unit normal vector $\mathbf{n} = \begin{pmatrix}\cos\theta_0 \\ \sin\theta_0\end{pmatrix}$, eccentricity e, and directrix distance parameter $c = -p$, substituting into (1.1.33.2)–(1.1.33.4) yields:

$$\mathbf{V} = \mathbf{I} - e^2\mathbf{n}\mathbf{n}^\top, \tag{1.1.34.3}$$

$$\mathbf{u} = -pe^2\mathbf{n}, \tag{1.1.34.4}$$

$$f = -p^2e^2 \tag{1.1.34.5}$$

Expressing $\mathbf{x}$ directly in polar coordinates yields

$$\mathbf{x} = r\mathbf{r}(\theta), \quad \text{where } \mathbf{r}(\theta) = \begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix} \tag{1.1.34.6}$$

Substituting into (1.1.34.2) yields

$$r^2\left(\mathbf{r}(\theta)^\top \mathbf{V}\mathbf{r}(\theta)\right) + 2r\left(\mathbf{u}^\top \mathbf{r}(\theta)\right) + f = 0 \tag{1.1.34.7}$$

Evaluating the inner products:

$$\mathbf{n}^\top \mathbf{r}(\theta) = \begin{pmatrix}\cos\theta_0 & \sin\theta_0\end{pmatrix}\begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix} = \cos(\theta - \theta_0) \tag{1.1.34.8}$$

$$\mathbf{r}(\theta)^\top \mathbf{V}\mathbf{r}(\theta) = \mathbf{r}(\theta)^\top\left(\mathbf{I} - e^2\mathbf{n}\mathbf{n}^\top\right)\mathbf{r}(\theta) = 1 - e^2\cos^2(\theta - \theta_0) \tag{1.1.34.9}$$

$$\mathbf{u}^\top \mathbf{r}(\theta) = -pe^2\cos(\theta - \theta_0) \tag{1.1.34.10}$$

Substituting these into (1.1.34.7):

$$r^2\left[1 - e^2\cos^2(\theta - \theta_0)\right] - 2rpe^2\cos(\theta - \theta_0) - p^2e^2 = 0 \tag{1.1.34.11}$$

Factoring the quadratic expression:

$$\left[r\left(1 + e\cos(\theta - \theta_0)\right) - ep\right]\left[r\left(1 - e\cos(\theta - \theta_0)\right) + ep\right] = 0 \tag{1.1.34.12}$$

Taking the positive real root for radial distance $r > 0$:

$$r\left(1 + e\cos(\theta - \theta_0)\right) = ep \tag{1.1.34.13}$$

yielding (1.1.34.1)

1.1.35 Gravitational parameter of a body μ is defined as

$$\mu = GM \tag{1.1.35.1}$$

where M is the mass of the body

1.1.36 Specific angular momentum for a satellite with position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$ is

$$h = \|\mathbf{r} \times \mathbf{v}\| \tag{1.1.36.1}$$

1.1.37 Specific Energy for a satellite with position vector $\mathbf{r}$, velocity vector $\mathbf{v}$ and gravitational parameter μ is

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} \tag{1.1.37.1}$$

where

$$r = \|\mathbf{r}\|, v = \|\mathbf{v}\| \tag{1.1.37.2}$$

1.1.38 The gravitational acceleration $\mathbf{a}$ acting on a satellite is given by Newton's law of gravitation:

$$\mathbf{a} = -\frac{\mu}{r^3}\mathbf{r} \tag{1.1.38.1}$$

1.1.39 The trajectory of a satellite under a central gravitational force is always a conic section.

Solution: Taking the cross product of $\mathbf{r}$ with $\mathbf{a}$:

$$\mathbf{r} \times \mathbf{a} = \mathbf{r} \times \left(-\frac{\mu}{r^3}\mathbf{r}\right) = \mathbf{0} \tag{1.1.39.1}$$

From (1.1.39.1)

$$\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \mathbf{v} \times \mathbf{v} + \mathbf{r} \times \mathbf{a} = \mathbf{0} \tag{1.1.39.2}$$

Hence, the specific angular momentum vector $\mathbf{h}$ is conserved:

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = \text{constant} \tag{1.1.39.3}$$

Crossing the acceleration $\mathbf{a}$ with $\mathbf{h}$:

$$\mathbf{a} \times \mathbf{h} = \left(-\frac{\mu}{r^3}\mathbf{r}\right) \times (\mathbf{r} \times \mathbf{v}) \tag{1.1.39.4}$$

$$= -\frac{\mu}{r^3}\left[(\mathbf{r} \cdot \mathbf{v})\,\mathbf{r} - (\mathbf{r} \cdot \mathbf{r})\,\mathbf{v}\right] \tag{1.1.39.5}$$

$$= \frac{\mu}{r^3}\left[r^2\mathbf{v} - (\mathbf{r} \cdot \mathbf{v})\,\mathbf{r}\right] \tag{1.1.39.6}$$

Using $\mathbf{r} \cdot \mathbf{v} = r\frac{dr}{dt}$, this simplifies to:

$$\mathbf{a} \times \mathbf{h} = \mu\left[\frac{\mathbf{v}}{r} - \frac{1}{r^2}\frac{dr}{dt}\mathbf{r}\right] = \mu\frac{d}{dt}\left(\frac{\mathbf{r}}{r}\right) \tag{1.1.39.7}$$

Integrating with respect to time t:

$$\mathbf{v} \times \mathbf{h} = \mu\left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.39.8}$$

where $\mathbf{C}$ is the integration constant. Taking the scalar product of (1.1.39.8) with $\mathbf{r}$:

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = \mu\mathbf{r} \cdot \left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.39.9}$$

$$\implies (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mu\left(r + rC\cos\theta\right) \tag{1.1.39.10}$$

$$\implies \|\mathbf{h}\|^2 = \mu r\left(1 + C\cos\theta\right) \tag{1.1.39.11}$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{C}$. Rearranging for r yields

$$r = \frac{\frac{\|\mathbf{h}\|^2}{\mu}}{1 + C\cos\theta} = \frac{ep}{1 + e\cos\theta} \tag{1.1.39.12}$$

This is the same equation as (1.1.34.1) where

$$ep = \frac{\|\mathbf{h}\|^2}{\mu} \tag{1.1.39.13}$$

$$\mathbf{e} = \mathbf{C} \tag{1.1.39.14}$$

1.1.40 The squared eccentricity e^2 of a satellite's trajectory is related to its specific orbital energy ε and specific angular momentum magnitude h by

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.40.1}$$

Solution: From (1.1.39.8) and (1.1.39.14),

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \tag{1.1.40.2}$$

Taking the magnitude squared of both sides using $\|\mathbf{e}\|^2 = \mathbf{e} \cdot \mathbf{e}$:

$$e^2 = \|\mathbf{e}\|^2 = \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \cdot \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \tag{1.1.40.3}$$

Expanding the inner product:

$$e^2 = \frac{\|\mathbf{v} \times \mathbf{h}\|^2}{\mu^2} - \frac{2}{\mu r}\left[\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h})\right] + \frac{\mathbf{r} \cdot \mathbf{r}}{r^2} \tag{1.1.40.4}$$

We know that

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mathbf{h} \cdot \mathbf{h} = h^2 \tag{1.1.40.5}$$

And,

$$\mathbf{r} \cdot \mathbf{r} = r^2 \tag{1.1.40.6}$$

Substituting these back into (1.1.40.4) yields

$$e^2 = \frac{v^2 h^2}{\mu^2} - \frac{2h^2}{\mu r} + 1 \tag{1.1.40.7}$$

$$= 1 + \frac{2h^2}{\mu^2}\left(\frac{v^2}{2} - \frac{\mu}{r}\right) \tag{1.1.40.8}$$

Substituting (1.1.37.1) into (1.1.40.8) yields:

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.40.9}$$

1.1.41 Consider the system A system comprising a bar, spring and mass is shown in the

figure Fig. 1.1.41.1 The bar, having negligible mass, is made of a material having Young's modulus E, cross-sectional area A and length L. The spring stiffness is k and the mass is M kg. Find the axial stiffness of the bar.

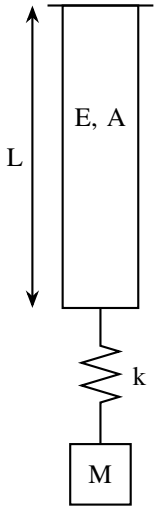


Fig. 1.1.41.1. Spring mass system

Solution: The axial stiffness of the bar is

$$k_b = \frac{AE}{L} \tag{1.1.41.1}$$

1.1.42 Find the overall spring constant in Fig. 1.1.41.1.

Solution: Using Hooke's law, by balancing the displacements,

$$\frac{Mg}{k_t} = \frac{Mg}{k_b} + \frac{Mg}{k} \tag{1.1.42.1}$$

$$\implies k_{eq} = \frac{k_b k}{k_b + k} \tag{1.1.42.2}$$

where k_t is the overall spring constant.

1.1.43 Show that the equation for motion for the mass in Fig. 1.1.43.1 is

$$\frac{d^2x}{dt^2} = -\frac{k}{M}x \tag{1.1.43.1}$$

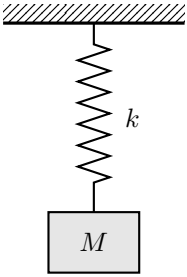


Fig. 1.1.43.1. Spring Mass system in vertical plane

Solution: When the mass M is at rest, let the system stretch by δ_{st}.Then,

$$Mg - k\delta_{st} = 0 \implies Mg = k\delta_{st} \tag{1.1.43.2}$$

When the mass is displaced downward by $x(t)$, using Hooke's law,

$$M\frac{d^2x}{dt^2} = Mg - k(\delta_{\text{st}} + x) \tag{1.1.43.3}$$

$$\tag{1.1.43.4}$$

yielding (1.1.43.1), upon substituting from (1.1.43.2).

2. BT 2025

A. Formulae

2.1.1 Exponential Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \tag{2.1.1.1}$$

2.1.2 The normal form of the equation of a line is

$$\mathbf{n}^\top \mathbf{x} = c \tag{2.1.2.1}$$

2.1.3 The parametric form of the equation of a line is

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m} \tag{2.1.3.1}$$

where

$$\mathbf{m}^\top \mathbf{n} = 0 \tag{2.1.3.2}$$

2.1.4 A circle with center $\mathbf{c}$ and radius r, can be expressed as (1.1.33.1) with

$$\mathbf{V} = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = -\mathbf{c}, \quad f = \|\mathbf{c}\|^2 - r^2 \tag{2.1.4.1}$$

Solution: The equation of a circle is

$$\|\mathbf{x}-\mathbf{c}\|^2 = r^2 \tag{2.1.4.2}$$

$$(\mathbf{x}-\mathbf{c})^\top (\mathbf{x}-\mathbf{c}) = r^2 \tag{2.1.4.3}$$

$$\mathbf{x}^\top\mathbf{x} - 2\mathbf{c}^\top\mathbf{x} + \mathbf{c}^\top\mathbf{c} - r^2 = 0 \tag{2.1.4.4}$$

yielding (2.1.4.1) upon comparing with (1.1.33.1)

2.1.5 (Line-Conic Intersection) The points of intersection of a line $L : \mathbf{x} = \mathbf{h}+\kappa\mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top\mathbf{V}\mathbf{x} + 2\mathbf{u}^\top\mathbf{x} + f = 0$ are given by

$$\mathbf{X} = \begin{pmatrix}\mathbf{h} & \mathbf{m}\end{pmatrix}\begin{pmatrix}1 & 1\\ \kappa_1 & \kappa_2\end{pmatrix} \tag{2.1.5.1}$$

where

$$\kappa_i = \frac{-\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u})\right]^2 - g(\mathbf{h})(\mathbf{m}^\top\mathbf{V}\mathbf{m})}}{\mathbf{m}^\top\mathbf{V}\mathbf{m}}, i = 1, 2. \tag{2.1.5.2}$$

with $g(\mathbf{h}) = \mathbf{h}^\top\mathbf{V}\mathbf{h} + 2\mathbf{u}^\top\mathbf{h} + f$.

2.1.6 (Intersection of Conics and Common Chord) The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ is given by the pencil

$$\mathbf{x}^\top(\mathbf{V}_1 + \mu\mathbf{V}_2)\mathbf{x} + 2(\mathbf{u}_1 + \mu\mathbf{u}_2)^\top\mathbf{x} + (f_1 + \mu f_2) = 0 \tag{2.1.6.1}$$

The value of μ corresponding to the degenerate conic (the radical axis / common chord line) is found from

$$\begin{vmatrix}\mathbf{V}_1 + \mu\mathbf{V}_2 & \mathbf{u}_1 + \mu\mathbf{u}_2\\ (\mathbf{u}_1 + \mu\mathbf{u}_2)^\top & f_1 + \mu f_2\end{vmatrix} = 0 \tag{2.1.6.2}$$

2.1.7 For a differentiable function $f(x)$,

$$f''(x) > 0 \quad \forall x \in (0,\infty) \implies f(x) \text{ is strictly convex} \tag{2.1.7.1}$$

2.1.8 (Gradient Descent Optimization) For the convex optimization problem

$$\min_x f(x), \tag{2.1.8.1}$$

$$x_{k+1} = x_k - \eta f'(x_k) \tag{2.1.8.2}$$

is the iterative parameter update with learning rate $\eta > 0$.

2.1.9 (Newton-Raphson Method) For a real-valued differentiable function $f(x)$, the root is iteratively approximated using its first-order Taylor series expansion about the current estimate x_n

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{2.1.9.1}$$

2.1.10 (System of Linear Equations) The matrix equation

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{2.1.10.1}$$

is consistent iff

$$\text{rank}\left(\mathbf{A}\right) = \text{rank}\left(\left[\mathbf{A} \mid \mathbf{b}\right]\right) = r \tag{2.1.10.2}$$

2.1.11 Sum of Independent Discrete Random Variables (Two Dice Roll)

- Let $X_1, X_2 \in \{1,2,3,4,5,6\}$ be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF)

$$p_{X_1}\left(n\right) = p_{X_2}\left(n\right) = \begin{cases} \frac{1}{6}, & 1 \leq n \leq 6 \\ 0, & \text{otherwise} \end{cases} \tag{2.1.11.1}$$

- (Convolution) Let $X = X_1 + X_2$ denote their sum. The PMF of X is given by the discrete convolution of $p_{X_1}\left(n\right)$ and $p_{X_2}\left(n\right)$:

$$p_X\left(n\right) = \Pr\left(X_1 + X_2 = n\right) = \sum_k \Pr\left(X_1 = n-k \mid X_2 = k\right) p_{X_2}\left(k\right) \tag{2.1.11.2}$$

$$= \sum_k p_{X_1}\left(n-k\right) p_{X_2}\left(k\right) = p_{X_1}\left(n\right) * p_{X_2}\left(n\right) \tag{2.1.11.3}$$

- Substituting (2.1.11.1) into (2.1.11.3) yields the triangular PMF

$$p_X\left(n\right) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \leq n \leq 7 \\ \frac{13-n}{36}, & 7 < n \leq 12 \\ 0, & n > 12 \end{cases} \tag{2.1.11.4}$$

The following code verifies (2.1.11.4) in Fig. 2.1.11.1.

codes/btq35/dice.py

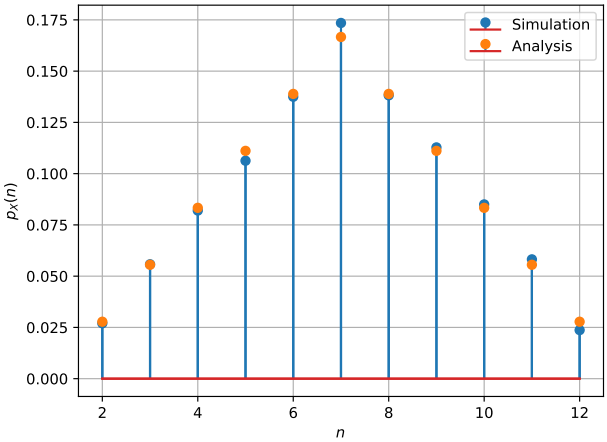


Fig. 2.1.11.1. PMF of the sum of two fair dice: Simulation vs Analysis

2.1.12 (Unit Step Response of an RC Circuit) Consider a series RC circuit excited by a step input $v_i(t) = V_0 u(t)$ where V_0 is a constant. Applying KVL around the loop,

$$RC\frac{dv_o}{dt} + v_o(t) = V_0 u(t) \tag{2.1.12.1}$$

yielding

$$\frac{V_0}{s} = sCRV_o(s) + V_o(s) \tag{2.1.12.2}$$

$$\implies V_o(s) = \frac{V_0}{s(RCs+1)} \tag{2.1.12.3}$$

$$= V_0\left[\frac{1}{s} - \frac{1}{s+\frac{1}{RC}}\right] \tag{2.1.12.4}$$

$$\implies v_o(t) = V_0\left(1 - e^{-\frac{t}{RC}}\right)u(t) \tag{2.1.12.5}$$

2.1.13 (Continuity) A real-valued function $f(x)$ is continuous at $x = x_0$ iff

$$\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x) = f(x_0) \tag{2.1.13.1}$$

2.1.14 (Differentiability) A function $f(x)$ is differentiable at $x = x_0$ if

$$f'(x_0^-) = \lim_{h\to 0^-}\frac{f(x_0+h)-f(x_0)}{h} = \lim_{h\to 0^+}\frac{f(x_0+h)-f(x_0)}{h} = f'(x_0^+) \tag{2.1.14.1}$$

2.1.15 If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$

$$\lim_{x \to x_0} [f(x) - f(x_0)] = \lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} (x - x_0) \right] = f'(x_0) \cdot 0 = 0 \tag{2.1.15.1}$$

2.1.16 Piecewise Boundary Conditions at $x = a$

- Consider a function $f(x)$ defined by differentiable branches $f_1(x)$ and $f_2(x)$ across a boundary point $x = a$:

$$f(x) = \begin{cases} f_1(x), & x \leq a \\ f_2(x), & x > a \end{cases} \tag{2.1.16.1}$$

For $f(x)$ to be continuous at $x = a$, the one-sided limits must equal the boundary value:

$$\lim_{x \to a^-} f_1(x) = \lim_{x \to a^+} f_2(x) \implies f_1(a) = f_2(a) \tag{2.1.16.2}$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives at the boundary must coincide:

$$f_1'(a) = f_2'(a) \tag{2.1.16.3}$$

2.1.17 The Z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \tag{2.1.17.1}$$

over the Region of Convergence (ROC) where the series converges absolutely.

2.1.18 The discrete unit step function is defined as

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{2.1.18.1}$$

and

$$u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}}, \quad |z| > 1 \tag{2.1.18.2}$$

2.1.19 Time-Shift Property

$$x(n - k) \overset{\mathcal{Z}}{\longleftrightarrow} z^{-k} X(z) \tag{2.1.19.1}$$

2.1.20 Standard Inverse Transform Pair: For any scalar constant α,

$$\alpha^n u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1 - \alpha z^{-1}}, \quad |z| > |\alpha| \tag{2.1.20.1}$$

2.1.21 (Trace of a Matrix) Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix with eigenvalues

$\lambda_1, \lambda_2, \ldots, \lambda_n$. Then,

$$\sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} a_{ii} = \mathrm{tr}\,(\mathbf{A}) \tag{2.1.21.1}$$

Solution: The characteristic polynomial of $\mathbf{A}$ is defined by

$$p_{\mathbf{A}}(\lambda) = \left|\lambda\mathbf{I} - \mathbf{A}\right| \tag{2.1.21.2}$$

$$= \prod_{i=1}^{n} (\lambda - \lambda_i) = \lambda^n - \left(\sum_{i=1}^{n} \lambda_i\right) \lambda^{n-1} + \cdots + (-1)^n \prod_{i=1}^{n} \lambda_i \tag{2.1.21.3}$$

Evaluating the determinant $\left|\lambda\mathbf{I} - \mathbf{A}\right|$ using cofactor expansion, the only term that contributes to λ^n and λ^{n-1} is the product of the principal diagonal entries

$$\prod_{i=1}^{n} (\lambda - a_{ii}) = \lambda^n - \left(\sum_{i=1}^{n} a_{ii}\right) \lambda^{n-1} + \ldots \tag{2.1.21.4}$$

where a_{ii} denotes the diagonal elements of $\mathbf{A}$. Equating the coefficient of λ^{n-1} from (2.1.21.3) and (2.1.21.4)

3. AG 2025

A. Formulae

3.1.1 (Bernoulli Distribution) For $X \in \{0, 1\}$, the pmf is defined as

$$P_X(k) = \begin{cases} p & k = 1 \\ q & k = 0 \\ 0 & \text{otherwise} \end{cases} \tag{3.1.1.1}$$

3.1.2 The moment generating function

$$\begin{aligned} M_X(z) \triangleq E\left[z^{-X}\right] &= p_X(1) z^{-1} + p_X(0) z^0 \\ &= p z^{-1} + q \end{aligned} \tag{3.1.2.1}$$

upon substituting from (2.1.17.1).

3.1.3 (Binomial Distribution) If X_i are independent, for

$$\begin{aligned} Y \triangleq \sum_{i=1}^{n} X_i, \quad M_Y(z) &= \prod_{i=1}^{n} M_{X_i}(z) \\ &= \left(pZ^{-1} + q\right)^n \end{aligned} \tag{3.1.3.1}$$

Solution: From (3.1.2.1),

$$M_Y(z) = E\left[z^{-\sum_{i=1}^{n} X_i}\right] = E\left[\prod_{i=1}^{n} z^{-X_i}\right] \tag{3.1.3.2}$$

$$= \prod_{i=1}^{n} E\left[z^{-X_i}\right] \tag{3.1.3.3}$$

yielding (3.1.3.1).

3.1.4 From (3.1.3.1),

$$\left(pz^{-1}+q\right)^n = {}^nC_k p^k q^{n-k} z^{-k} \tag{3.1.4.1}$$

$$\implies p_Y(k) = \begin{cases} {}^nC_k p^k q^{n-k} & 0 \le k \le n \\ 0 & \text{otherwise} \end{cases} \tag{3.1.4.2}$$

using the definition in (3.1.2.1) along with (2.1.17.1).

3.1.5

$$\begin{aligned} \sinh(at)\,u(t) &\xleftrightarrow{\mathcal{L}} \frac{a}{s^2-a^2} \\ \cosh(at)\,u(t) &\xleftrightarrow{\mathcal{L}} \frac{s}{s^2-a^2} \end{aligned} \tag{3.1.5.1}$$

Solution: From (1.1.13.1),

$$\cos(\jmath at)u(t) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2+(\jmath a)^2} \tag{3.1.5.2}$$

yielding (3.1.5.1).

3.1.6 (Geometric Progression)

$$\sum_{i=0}^{n-1} q^i = \frac{q^n-1}{q-1} \tag{3.1.6.1}$$

3.1.7

$$\begin{aligned} \sin(x) &= \sum_{i=1}^{n} -1^{i+1} \frac{x^{2i-1}}{(2i-1)!} \\ \cos(x) &= \sum_{i=1}^{n} -1^{i+1} \frac{x^{i-1}}{i-1!} \end{aligned} \tag{3.1.7.1}$$

3.1.8 If

$$f(\alpha x, \alpha y) = \alpha^n \alpha^n f(x,y), \tag{3.1.8.1}$$

$f(x,y)$ is homogeneous of degree n. Then,

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = nf \tag{3.1.8.2}$$

4. GATE CE-2 2025

A. Formulae

4.1.1 (Integration by Parts) For differentiable functions $u(x)$ and $v(x)$, the integration by parts formula is:

$$\int u\,dv = uv - \int v\,du \tag{4.1.1.1}$$

4.1.2 Let

$$\mathbf{V} = \begin{pmatrix} u \\ v \end{pmatrix}, \nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix}, \tag{4.1.2.1}$$

(4.1.2.2)

The curl and divergence of $\mathbf{V}$ are then defined respectively as

$$\nabla \times \mathbf{V} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \mathbf{e}_3 \tag{4.1.2.3}$$

$$\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}. \tag{4.1.2.4}$$

4.1.3 If $B \subset A, AB = 1$.

4.1.4 (Conditional Probability)

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \quad \Pr(B) > 0. \tag{4.1.4.1}$$

4.1.5 Mean and Variance of Discrete Random Variables:

$$\mu = E[X] = \sum x_i p_i \tag{4.1.5.1}$$

$$\sigma^2 = E[X^2] - (E[X])^2 = \sum x_i^2 p_i - \mu^2 \tag{4.1.5.2}$$

B. Problems

4.2.1 An electricity utility company charges ₹7 per kWh (kilo watt-hour). If a 40-watt desk light is left on for 10 hours each night for 180 days, what would be the cost of energy consumption? If the desk light is on for 2 more hours each night for the 180 days, what would be the percentage-increase in the cost of energy consumption? (CE2 2025)

a) ₹604.8; 10% b) ₹504; 20% c) ₹604.8; 12% d) ₹720; 15%

Solution: The following code generates the result.

codes/ce2q3/ce2q3.py

4.2.2 A bag contains Violet (V), Yellow (Y), Red (R), and Green (G) balls. On counting them, the following results are obtained:

a) The sum of Yellow balls and twice the number of Violet balls is 50.
b) The sum of Violet and Green balls is 50.
c) The sum of Yellow and Red balls is 50.
d) The sum of Violet and twice the number of Red balls is 50.

Which one of the following correctly represents the balls in the bag? (CE2 2025)

a) $V = 10\%$, $R = 20\%$, $Y = 30\%$, $G = 40\%$
b) $V = 40\%$, $R = 10\%$, $Y = 20\%$, $G = 30\%$
c) $V = 30\%$, $R = 40\%$, $Y = 10\%$, $G = 10\%$
d) $V = 20\%$, $R = 30\%$, $Y = 40\%$, $G = 10\%$

Solution: Let the numbers of Violet, Yellow, Red, and Green balls be represented by V, Y, R, G, respectively. The four conditions give

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \tag{4.2.2.1}$$

where

$$\mathbf{x} = \begin{pmatrix} V \\ Y \\ R \\ G \end{pmatrix}, \qquad \mathbf{A} = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 50 \\ 50 \\ 50 \\ 50 \end{pmatrix}. \tag{4.2.2.2}$$

Using row operations,

$$\begin{pmatrix} 2 & 1 & 0 & 0 & 50 \\ 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 1 & 0 & 50 \\ 1 & 0 & 2 & 0 & 50 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 2 & 1 & 0 & 0 & 50 \\ 0 & 1 & 1 & 0 & 50 \\ 1 & 0 & 2 & 0 & 50 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 0 & -2 & -50 \\ 0 & 1 & 1 & 0 & 50 \\ 0 & 0 & 2 & -1 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 0 & -2 & -50 \\ 0 & 0 & 1 & 2 & 100 \\ 0 & 0 & 2 & -1 & 0 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 0 & -2 & -50 \\ 0 & 0 & 1 & 2 & 100 \\ 0 & 0 & 0 & -5 & -200 \end{pmatrix} \xrightarrow[\substack{R_1 \leftarrow R_1 - R_4 \\ R_2 \leftarrow R_2 + 2R_4 \\ R_3 \leftarrow R_3 - 2R_4}]{R_4 \leftarrow -\frac{1}{5} R_4} \begin{pmatrix} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 0 & 30 \\ 0 & 0 & 1 & 0 & 20 \\ 0 & 0 & 0 & 1 & 40 \end{pmatrix}.$$

Hence,

$$V = 10, \qquad Y = 30, \qquad R = 20, \qquad G = 40. \tag{4.2.2.3}$$

The following code generates Fig. 4.2.2.1.

codes/ce2q5/ce2q5.py

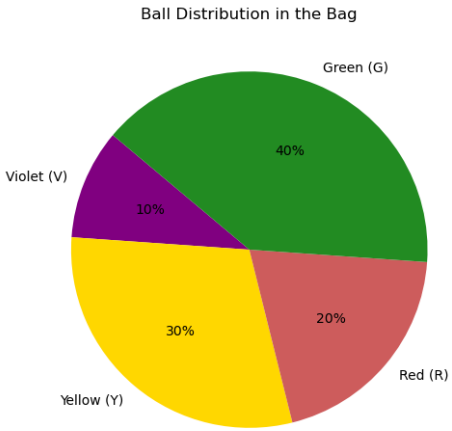


Fig. 4.2.2.1. Pie chart representing ball distribution.

4.2.3 Consider a five-digit number PQRST that has distinct digits P, Q, R, S, and T, and satisfies the following conditions: $P < Q$, $S > P > T$, and $R < T$. If integers 1 through 5 are used to construct such a number, the value of P is: (CE2 2025)

a) 1 b) 2 c) 3 d) 4

Solution: Using the following code, $P = 3$.

codes/ce2q8/ce2q8.c

4.2.4 A business person buys potatoes of two different varieties P and Q, mixes them in a certain ratio and sells them at ₹192 per kg. The cost of variety P is ₹800 for 5 kg. The cost of variety Q is ₹800 for 4 kg. If the person gets 8% profit, what is the $P : Q$ ratio (by weight)? (CE2 2025)

a) $5 : 4$ b) $3 : 4$ c) $3 : 2$ d) $1 : 1$

Solution: From the following code, the required ratio is $5 : 4$.

codes/ce2q9/ce2q9.c

4.2.5 Three villages P, Q, and R are located in such a way that the distance $PQ = 13$ km, $QR = 14$ km, and $RP = 15$ km. A straight road joins Q and R. It is proposed to connect P to this road QR by constructing another road. What is the minimum possible length (in km) of this connecting road? (CE2 2025)

a) 10.5 b) 11.0 c) 12.0 d) 12.5

Solution: The following code

codes/ce2q10/ce2q10.py

generates Fig. 4.2.5.1.

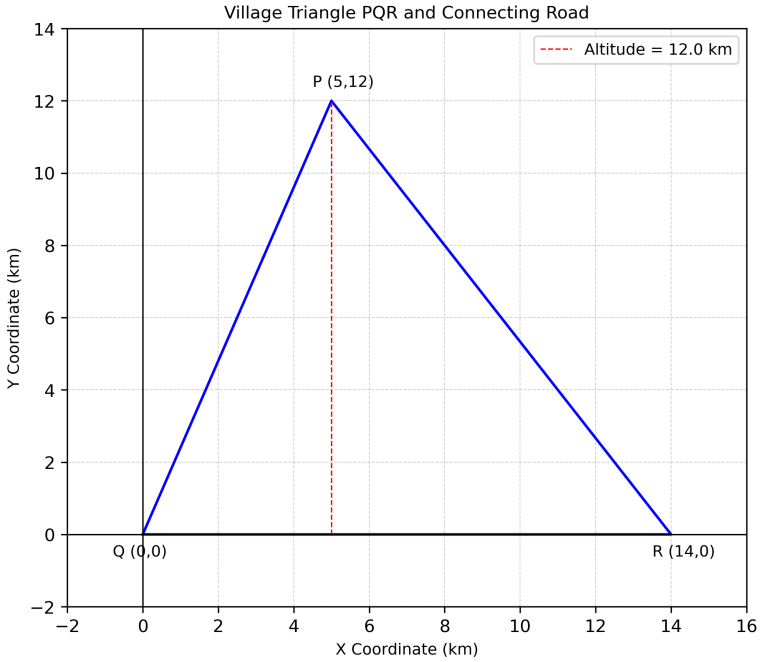


Fig. 4.2.5.1. Triangle PQR geometry and altitude from P to QR.

4.2.6 For the matrix $\mathbf{A}$ given below, the transpose is

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 4 & 5 \\ 4 & 3 & 2 \end{pmatrix} \tag{4.2.6.1}$$

(CE2 2025)

a) $\begin{pmatrix} 2 & 1 & 4 \\ 3 & 4 & 3 \\ 4 & 5 & 2 \end{pmatrix}$ b) $\begin{pmatrix} 4 & 3 & 2 \\ 5 & 4 & 1 \\ 2 & 3 & 4 \end{pmatrix}$ c) $\begin{pmatrix} 4 & 2 & 3 \\ 5 & 1 & 4 \\ 2 & 4 & 3 \end{pmatrix}$ d) $\begin{pmatrix} 2 & 3 & 4 \\ 1 & 4 & 5 \\ 4 & 3 & 2 \end{pmatrix}$

Solution: Using the following code,

$$\mathbf{A}^\top = \begin{pmatrix} 2 & 1 & 4 \\ 3 & 4 & 3 \\ 4 & 5 & 2 \end{pmatrix}. \tag{4.2.6.2}$$

codes/ce2q11/ce2q11.py

4.2.7 Integration of $\ln(x)$ with respect to x, i.e., $\int \ln(x)\, dx$ =_____
(CE2 2025)

a) $x\ln(x) - x + C$
b) $x - \ln(x) + C$
c) $x\ln(x) + x + C$
d) $\ln(x) - x + C$

Solution: From (4.1.1.1), letting

$$u = \ln(x), dv = dx, \tag{4.2.7.1}$$

$$du = \frac{1}{x}, v = x \tag{4.2.7.2}$$

$$\implies \int \ln(x)\, dx = x\ln(x) - \int 1\, dx \tag{4.2.7.3}$$

$$= x\ln(x) - x + C. \tag{4.2.7.4}$$

4.2.8 Consider a velocity vector $\mathbf{V}$ in (x, y, z) coordinates given by $\mathbf{V} = u\mathbf{x} + v\mathbf{y}$. Pick one or more CORRECT statement(s) from the choices given below. (CE2 2025)

a) z-component of Curl of velocity: $\nabla \times \mathbf{V} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\mathbf{z}$
b) z-component of Curl of velocity: $\nabla \times \mathbf{V} = \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right)\mathbf{z}$
c) Divergence of velocity: $\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$
d) Divergence of velocity: $\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$

Solution: See (4.1.2.3) and (4.1.2.4).

4.2.9 Given that A and B are not null sets, which of the following statements regarding probability is/are CORRECT? (CE2 2025)

a) $\Pr(A \cap B) = \Pr(A)\Pr(B)$ if A and B are mutually exclusive.
b) Conditional probability, $\Pr(A \mid B) = 1$ if $B \subset A$.
c) $\Pr(A \cup B) = \Pr(A) + \Pr(B)$, if A and B are mutually exclusive.
d) $\Pr(A \cap B) = 0$, if A and B are independent.

Solution:

a) Since $B \subset A$, from Definition (4.1.3) and (4.1.4.1),

$$AB = B, \tag{4.2.9.1}$$

$$\Pr(A \mid B) = \frac{\Pr(AB)}{\Pr(B)} = \frac{\Pr(B)}{\Pr(B)} = 1. \tag{4.2.9.2}$$

b) From (1.1.3.1),

$$\Pr(A + B) = \Pr(A) + \Pr(B). \tag{4.2.9.3}$$

Other statements are incorrect.

4.2.10 Pick the CORRECT solution for the following differential equation

$$\frac{dy}{dx} = e^{x-y}. \tag{4.2.10.1}$$

(CE2 2025)

a) $y = \ln\left(e^x + \text{Constant}\right)$
b) $\ln(y) = x + \text{Constant}$
c) $\ln(y) = \ln\left(e^x\right) + \text{Constant}$
d) $y = x + \text{Constant}$

Solution: The nonlinear equation is transformed using $u = e^y$.

$$\text{Given:} \quad \frac{dy}{dx} = e^{x-y} = e^x e^{-y}, \tag{4.2.10.2}$$

$$\text{Let:} \quad u = e^y, \tag{4.2.10.3}$$

$$\frac{du}{dx} = e^y \frac{dy}{dx} = e^x, \tag{4.2.10.4}$$

$$\text{Laplace transform:} \quad \mathcal{L}\left\{\frac{du}{dx}\right\} = sU(s) - u(0), \tag{4.2.10.5}$$

$$\mathcal{L}\{e^x\} = \frac{1}{s-1}, \tag{4.2.10.6}$$

$$sU(s) - u(0) = \frac{1}{s-1}, \tag{4.2.10.7}$$

$$U(s) = \frac{1}{s(s-1)} + \frac{u(0)}{s}, \tag{4.2.10.8}$$

$$\tag{4.2.10.9}$$

$$\text{Partial fractions:} \quad \frac{1}{s(s-1)} = -\frac{1}{s} + \frac{1}{s-1}, \tag{4.2.10.10}$$

$$U(s) = \frac{u(0)-1}{s} + \frac{1}{s-1}, \tag{4.2.10.11}$$

$$\text{Inverse Laplace:} \quad u(x) = u(0) - 1 + e^x, \tag{4.2.10.12}$$

$$\text{Since } u = e^y: \quad e^y = e^x + C, \tag{4.2.10.13}$$

$$\text{Taking Log:} \quad y = \ln\left(e^x + C\right). \tag{4.2.10.14}$$

Hence, the correct solution is

$$y = \ln\left(e^x + \text{Constant}\right). \tag{4.2.10.15}$$

Therefore, the correct option is (A).

codes/ce2q36/ce2q36.py

4.2.11 Consider the function given below and pick one or more CORRECT statement(s) from the following choices.

$$f(x) = x^3 - \frac{15}{2}x^2 + 18x + 20. \tag{4.2.11.1}$$

(CE2 2025)

a) $f(x)$ has a local minimum at $x = 3$.
b) $f(x)$ has a local maximum at $x = 3$.
c) $f(x)$ has a local minimum at $x = 2$.
d) $f(x)$ has a local maximum at $x = 2$.

Solution: The critical points are obtained from $f'(x) = 0$:

$$f'(x) = 3x^2 - 15x + 18 \tag{4.2.11.2}$$
$$= 3(x^2 - 5x + 6) \tag{4.2.11.3}$$
$$= 3(x-2)(x-3). \tag{4.2.11.4}$$

Hence,

$$x = 2, \qquad x = 3. \tag{4.2.11.5}$$

The second derivative is

$$f''(x) = 6x - 15. \tag{4.2.11.6}$$

At $x = 2$,

$$f''(2) = 12 - 15 = -3 < 0, \tag{4.2.11.7}$$

so $x = 2$ is a local maximum.
At $x = 3$,

$$f''(3) = 18 - 15 = 3 > 0, \tag{4.2.11.8}$$

so $x = 3$ is a local minimum.
Therefore, the correct statements are (A) and (D).

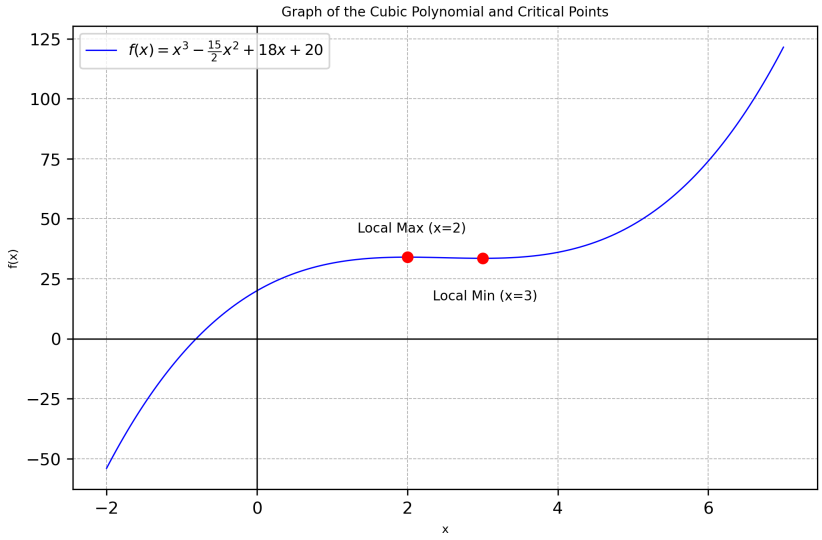


Fig. 4.2.11.1. Graph of the cubic function.

codes/ce2q44/ce2q44.py

4.2.12 Pick the CORRECT eigen value(s) of the matrix $\mathbf{A}$ from the following choices.

$$\mathbf{A} = \begin{pmatrix} 6 & 8 \\ 4 & 2 \end{pmatrix}. \tag{4.2.12.1}$$

(CE2 2025)

a) 10 b) 4 c) -2 d) -10

Solution: The eigenvalues satisfy the characteristic equation

$$\det(\mathbf{A} - \lambda I) = 0 \tag{4.2.12.2}$$

$$\det \begin{pmatrix} 6-\lambda & 8 \\ 4 & 2-\lambda \end{pmatrix} = 0 \tag{4.2.12.3}$$

$$(6-\lambda)(2-\lambda) - 32 = 0 \tag{4.2.12.4}$$

$$\lambda^2 - 8\lambda - 20 = 0 \tag{4.2.12.5}$$

$$(\lambda - 10)(\lambda + 2) = 0. \tag{4.2.12.6}$$

Hence,

$$\lambda = 10, \qquad \lambda = -2. \tag{4.2.12.7}$$

So, the correct options are (A) and (C).

codes/ce2q45/ce2q45.py

4.2.13 In the pin-jointed truss shown in the figure, the members that carry zero force are identified. Which of the following options is/are zero-force members?

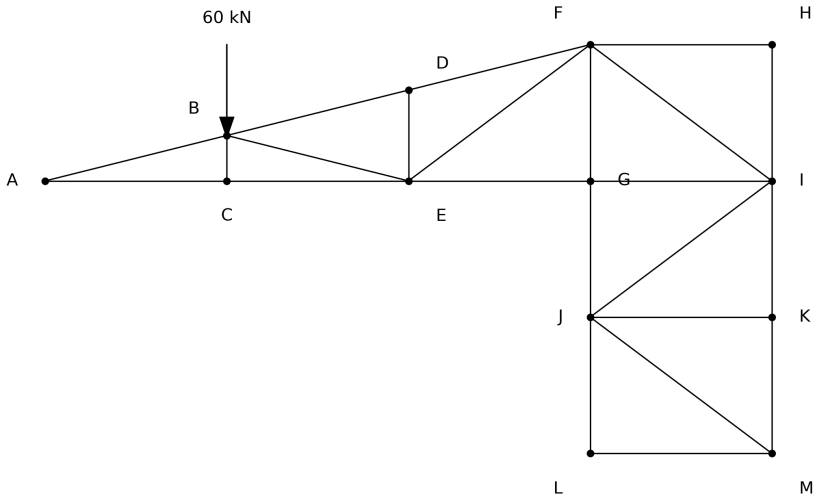


Fig. 4.2.13.1. Truss diagram for identifying zero-force members in Q46.

a) BC b) EG c) FI d) JK

Solution: We apply the standard zero-force member rules for pin-jointed trusses:

a) **At Joint C:** The bottom chord members AC and CE are collinear, and no external load acts at joint C. Therefore, member BC carries zero force:

$$BC = 0 \tag{4.2.13.1}$$

b) **At Joint J:** The horizontal chord members meeting at joint J are collinear, and no external load or support reaction is applied at joint J. Hence, member JK carries zero force:

$$JK = 0 \tag{4.2.13.2}$$

Thus, among the choices provided, both BC and JK are zero-force members. Therefore, the correct options are (A) and (D).

codes/ce2q46/ce2q46.py

4.2.14 A compound has a general formula $C_aH_bO_cN_d$ and molecular weight 187. A 935 mg/l solution of the compound is prepared in distilled deionized water. The Total Organic Carbon (TOC) is measured as 360 mg/l (as C). The Chemical Oxygen Demand (COD) and the Total Kjeldahl Nitrogen (TKN) are determined as 600 mg/l (as O_2) and 140 mg/l (as N), respectively.
As per the reaction

$$C_aH_bO_cN_d + \left(\frac{4a+b-2c-3d}{4}\right) O_2 \rightarrow aCO_2 + \frac{b-3d}{2} H_2O + dNH_3, \tag{4.2.14.1}$$

which of the following options is/are CORRECT?
Atomic weights: $C = 12$, $H = 1$, $O = 16$, $N = 14$

Quantity	Value
Molecular Weight	187
Solution Concentration	935
TOC	360
COD	600
TKN	140

TABLE 4.2.14
GIVEN VALUES

tables/ce2q49/ce2q49.tex

a) $a = 6$ b) $b = 7$ c) $c = 5$ d) $d = 3$

Solution: The concentration of the compound is

$$C_{\text{soln}} = 935 \text{ mg/l} = 0.935 \text{ g/l}, \tag{4.2.14.2}$$

so the molar concentration is

$$C_m = \frac{0.935}{187} = 5 \times 10^{-3} \text{ mol/l}. \tag{4.2.14.3}$$

From the TOC data,

$$\left(5 \times 10^{-3}\right)(12a) = 360 \times 10^{-3}, \tag{4.2.14.4}$$

$$0.06a = 0.36, \tag{4.2.14.5}$$

$$a = 6. \tag{4.2.14.6}$$

From the TKN data,

$$\left(5 \times 10^{-3}\right)(14d) = 140 \times 10^{-3}, \tag{4.2.14.7}$$

$$0.07d = 0.14, \tag{4.2.14.8}$$

$$d = 2. \tag{4.2.14.9}$$

Hence, option (D) is false.
Using the molecular weight,

$$12a + b + 16c + 14d = 187, \tag{4.2.14.10}$$

$$72 + b + 16c + 28 = 187, \tag{4.2.14.11}$$

$$b + 16c = 87. \tag{4.2.14.12}$$

From the COD data,

$$C_{O_2} = 600 \text{ mg/l} = 0.6 \text{ g/l}, \tag{4.2.14.13}$$

and the oxygen consumed per mole of compound is

$$\frac{0.6}{5 \times 10^{-3}} = 120 \text{ g/mol of } O_2, \tag{4.2.14.14}$$

$$n_{O_2} = \frac{120}{32} = 3.75. \tag{4.2.14.15}$$

Therefore,

$$\frac{4a + b - 2c - 3d}{4} = 3.75, \tag{4.2.14.16}$$

$$18 + b - 2c = 15, \tag{4.2.14.17}$$

$$b - 2c = -3. \tag{4.2.14.18}$$

Thus,

$$\begin{pmatrix} 1 & 16 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} b \\ c \end{pmatrix} = \begin{pmatrix} 87 \\ -3 \end{pmatrix}. \tag{4.2.14.19}$$

Row reduction gives

$$b = 7, \qquad c = 5, \tag{4.2.14.20}$$

$$a = 6, \qquad d = 2. \tag{4.2.14.21}$$

Hence,

$$C_a H_b O_c N_d = C_6 H_7 O_5 N_2. \tag{4.2.14.22}$$

Therefore, the correct options are (A), (B), and (C).

codes/ce2q49/ce2q49.py

4.2.15 Consider a discrete random variable X whose probabilities are given below. The standard deviation of the random variable is _____ (round off to one decimal place).

xi	1	2	4	8
P(X = xi)	0.3	1	0.3	0.3

TABLE 4.2.15
PROBABILITY TABLE

tables/ce2q51/ce2q51.tex

(CE2 2025)

Solution: The mean is

$$\mu = E[X] = \sum_i x_i p_i \tag{4.2.15.1}$$

$$= 1(0.3) + 2(0.1) + 4(0.3) + 8(0.3) = 4.1. \tag{4.2.15.2}$$

The second moment is

$$E[X^2] = \sum_i x_i^2 p_i \tag{4.2.15.3}$$

$$= 1^2(0.3) + 2^2(0.1) + 4^2(0.3) + 8^2(0.3) = 24.7. \tag{4.2.15.4}$$

Hence,

$$\sigma^2 = E[X^2] - \mu^2 = 24.7 - (4.1)^2 = 7.89, \tag{4.2.15.5}$$

and therefore

$$\sigma = \sqrt{7.89} \approx 2.8. \tag{4.2.15.6}$$

codes/ce2q51/ce2q51.py

4.2.16 The figure shows a network diagram for a construction project. The activities A, B, C, D, E, and F are represented by arrows and their durations are shown in the figure.

The total float available for the activity E in day(s) is equal to _____ (round off to the nearest integer). (CE2 2025)

Solution: Let x_i denote the occurrence time of event i and define

$$\mathbf{x} = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{pmatrix}^\top. \tag{4.2.16.1}$$

For an activity from event i to event j with duration d_{ij}, the precedence constraint is

$$x_j - x_i \geq d_{ij}. \tag{4.2.16.2}$$

For the given network,

$$x_2 - x_1 \geq 3, \qquad x_3 - x_1 \geq 5, \tag{4.2.16.3}$$

$$x_4 - x_2 \geq 5, \qquad x_4 - x_3 \geq 2, \tag{4.2.16.4}$$

$$x_5 - x_4 \geq 4, \qquad x_5 - x_3 \geq 8. \tag{4.2.16.5}$$

These constraints can be written in the matrix form

$$\mathbf{B}\mathbf{x} \geq \mathbf{d}, \tag{4.2.16.6}$$

where

$$\mathbf{B} = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 0 & 1 \end{pmatrix}, \qquad \mathbf{d} = \begin{pmatrix} 3 & 5 & 5 & 2 & 4 & 8 \end{pmatrix}^\top. \tag{4.2.16.7}$$

The project starts at event 1, hence $x_1 = 0$.

The forward-time optimization problem is therefore

$$\begin{aligned} \underset{\mathbf{x}}{\text{minimize}} \quad & x_5 \\ \text{subject to} \quad & \mathbf{B}\mathbf{x} \geq \mathbf{d}, \\ & x_1 = 0. \end{aligned} \tag{4.2.16.8}$$

The optimal solution is

$$\mathbf{x}^E = \begin{pmatrix} 0 & 3 & 5 & 8 & 13 \end{pmatrix}^\top, \tag{4.2.16.9}$$

so that $E_4 = 8$ and $T = E_5 = 13$ days.

For the backward pass, the latest feasible occurrence time of event 4 is

$$x_4^L = 9. \tag{4.2.16.10}$$

For activity $E : 4 \to 5$, with duration $d_E = 4$ days, the latest start time is

$$LST_E = L_5 - d_E = 13 - 4 = 9. \tag{4.2.16.11}$$

Therefore, the total float is

$$TF_E = LST_E - EST_E = 9 - 8 = 1 \text{ day}. \tag{4.2.16.12}$$

Hence, the total float available for activity E is

$$\boxed{1}. \tag{4.2.16.13}$$

codes/ce2q53/ce2q53.py